

HAT-P-32b

R-0037

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_171205 · created 2026-07-09T07:58:21+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458092.6592 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.053
Transit depth (Rp/Rs)^2	3.45 +/- 1.98 [%]
Orbital Inclination (inc)	84.1 +/- 1.72
Airmass coefficient 1 (a1)	1.0 +/- 0.12
Airmass coefficient 2 (a2)	-0.0 +/- 0.27
Scatter in the residuals of the lightcurve fit is	12.47 %
Best Comparison Star	#1 - [607, 97]
Optimal Aperture	3.02
Optimal Annulus	8.15
Transit Duration (day)	0.1227 +/- 0.0098

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.186 $\pm$ 0.053 vs 0.1489 (deviation 0.67 σ)
Duration fit vs published	0.1227 d vs 0.1304 d (deviation 0.75 σ)
Mid-time O-C	-2.6 min
Depth SNR	3.3
Residual scatter	12.47 %

Light curve

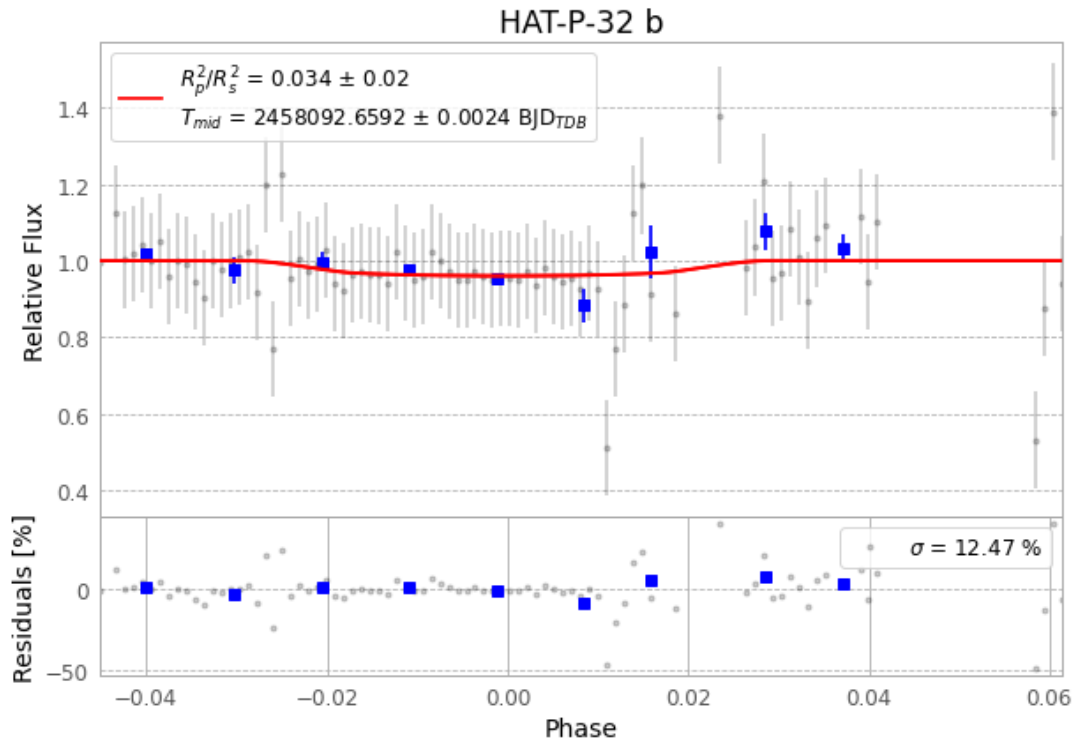


Figure 1 — FinalLightCurve_HAT-P-32 b_2017-12-04.png (SHA-256 5c082d06661b99b7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 256], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ab5fd2d17861bad2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

109 FITS frames (72 MB), HATP-32171205012142.FITS → HATP-32171205065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171205.log (91563cc2a398...), FinalLightCurve_HAT-P-32 b_2017-12-04.png (5c082d06661b...), FinalLightCurve_HAT-P-32 b_2017-12-04.csv (314f0563d066...)

Compute resources

Wall time 135.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.